ASSIGNMENT - 01

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MACHINE LEARNING.

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GUIDED BY: Dr. K. SIVAKUMAR.

1. PROBLEM : Identify Regions where people are at high Risk of poverty.

Input data : Income, Employment, Education & Population.

ALGORITHM : Decision Tree.

WORKING :

Socio Economic Data

↓

Data Preprocessing.

↓

Feature Selection.

↓

Decision Tree.

↓

Classification.

ADVANTAGES :- • Easy to Understand.

LIMITATIONS : • Poor Quality

• Incomplete Socio-Economic.

02.

PROBLEM: Automatically Identify People eligible for government welfare schemes.

FEATURES: Income, Family Size, Status & Housing.

ALGORITHM:

Applicant Data

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Data Cleaning

↓

Feature Selection

↓

Eligiblity Precision.

ADVANTAGES: • Reduces Manual Work.

DISADVANTAGES: It may produce under Predictions.

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03. PROBLEM:- Predict crop yield wing Historical data

FEATURES: RAINFALL, TEMPARATURE, Soil type, fertilizer, Irrigation

ALGORITHM:

Historical Agriculture data

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Data pre-processing.

↓

Feature Selection.

↓

Decision - Tree Regression.

↓

Estimated Crop yield.

ADVANTAGES: Supports Agriculture planning.

LIMITATIONS: Unexpected Whether Conditions.

04.

PROBLEM: Detect disease from crop Images.

INPUT: Images of Healthy & diseased leaves

ALGORITHM:-

Leaf Image → Preprocessing → Extraction →

Classification. ← ML Classifier ←

ADVANTAGES : • Helps the Farmer.

LIMITATIONS : • Image Quality affect prediction.

05.

PROBLEM: Predict product demand to prevent over stocking.

FEATURES.: Sales History, Season, Price

ALGORITHM:

Sales

↓

Pre-processing

↓

ML-Model

↓

Stock Planning.

ADVANTAGES : • Reduces food.

LIMITATIONS : Sudden changes in Demand.